

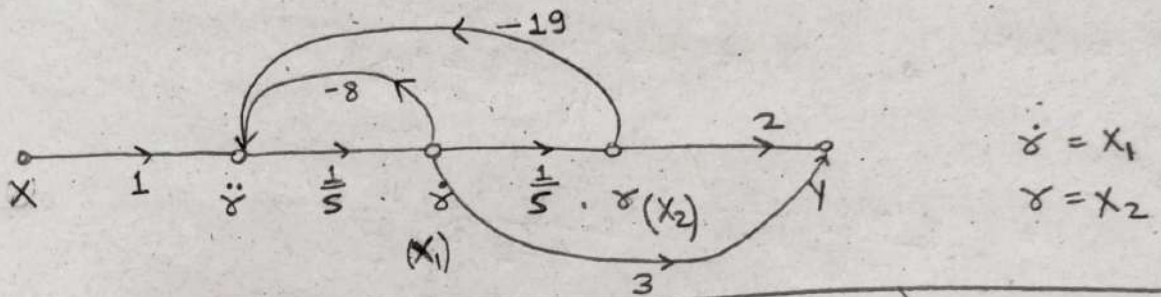
① $\frac{Y(s)}{X(s)} = \frac{3s+2}{(s+4)^2 + 3} = \frac{3s+2}{s^2 + 8s + 19} = \underbrace{(3s+2)}_{G_1} \underbrace{\left(\frac{1}{s^2 + 8s + 19} \right)}_{G_2}$

$$G_2 = \frac{R}{X}$$

$$G_1 = \frac{Y}{R}$$

$$Y = G_1 R = (3s+2) R$$

$$\frac{R}{X} = \frac{1}{s^2 + 8s + 19} \Rightarrow X = (s^2 + 8s + 19) R$$



$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -8 & -19 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u$$

$$y = \begin{pmatrix} 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} u$$

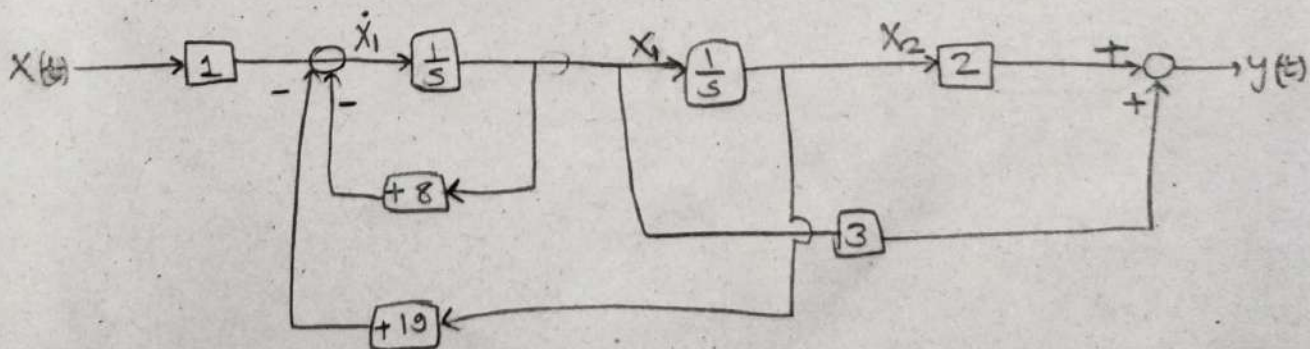
Controller

Canonical

Form $\rightarrow$

State Space

Representation



Impulse Response $\Rightarrow$

$$X(s) = \delta(s) \quad L(s(t)) = 1$$

$$\Rightarrow Y(s) = L^{-1}\left(\frac{3s+2}{(s+4)^2+3}\right) = L^{-1}\left(\frac{3(s+4)}{(s+4)^2+3} + \frac{-10}{(s+4)^2+3}\right)$$

$$\Rightarrow y(t) = 3e^{-4t} \cos(\sqrt{3}t) - \frac{10}{\sqrt{3}} e^{-4t} \sin(\sqrt{3}t)$$

Step Response $\Rightarrow$

$$X(s) = U(s) \Rightarrow Y(s) = L^{-1}\left(\frac{(3s+2)}{s((s+4)^2+3)}\right)$$

$$= L^{-1}\left(\frac{\frac{2}{19s}}{1} + \frac{(-2/19)s + (41/19)}{(s+4)^2+3}\right)$$

$$= L^{-1}\left(\frac{\frac{2}{19s}}{1} + \frac{(-2/19)(s+4)}{(s+4)^2+3} + \frac{(49/19)}{(s+4)^2+3}\right)$$

$$\Rightarrow y(t) = \frac{2}{19} - \frac{2}{19} e^{-4t} \cos \sqrt{3}t + \frac{49}{19\sqrt{3}} e^{-4t} \sin \sqrt{3}t$$

**

② $T(s) = \frac{Y(s)}{R(s)}$

For a step input $\Rightarrow$

(a) $1.05 \leq 50\%$

2.1. settling criterion

(b) $T_s < 4s$

(c) $T_p < 1 \text{ sec}$

$$T_s = \frac{4}{\zeta \omega_n}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

$$\zeta = \frac{\ln(1.05/100)}{\sqrt{\pi^2 + \ln^2(1.05/100)}}$$

let $T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

Now $T_s = \frac{4}{\zeta \omega_n} < 4 \Rightarrow 1 < \zeta \omega_n$
or $\omega_n \cos \theta > 1$

$T_p < 1 \Rightarrow \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} < 1 \Rightarrow \omega_n \sqrt{1-\zeta^2} > \pi$
 $\Rightarrow \omega_n \sin \theta > \pi$

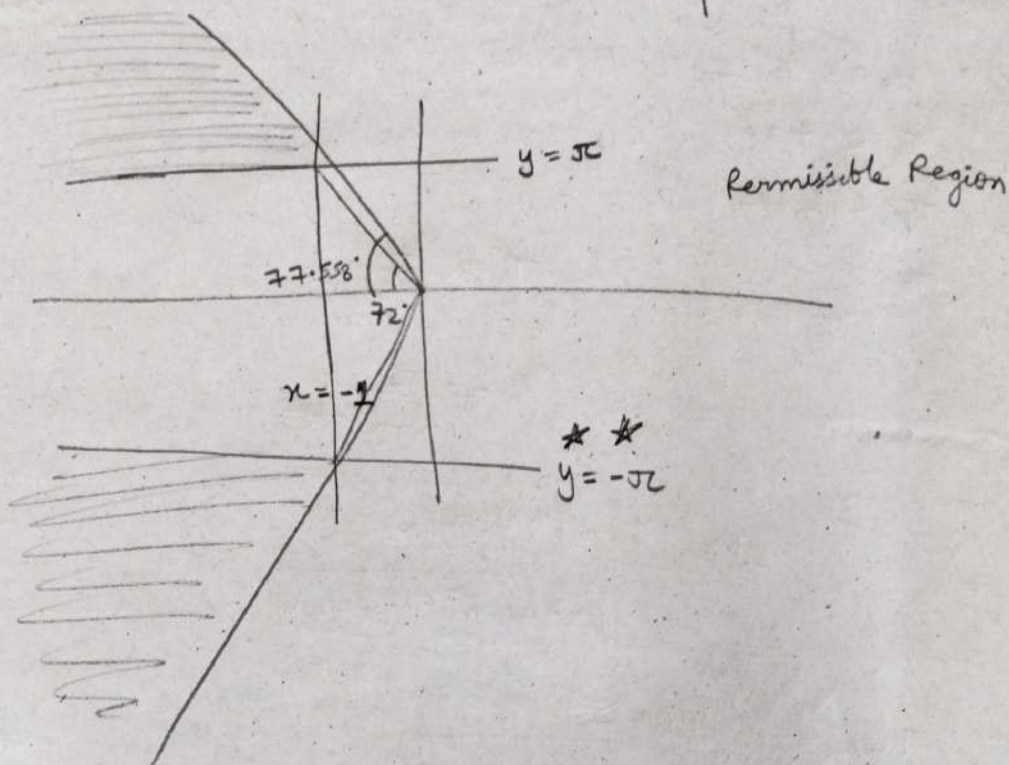
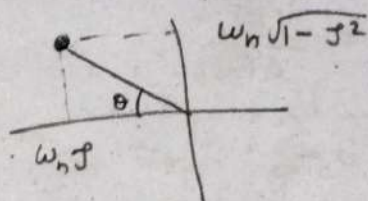
$1.05 \leq 50\% \Rightarrow \zeta \leq \frac{\left| \ln \left(\frac{50}{100} \right) \right|}{\sqrt{\pi^2 + \ln^2 \left(\frac{50}{100} \right)}} \leq \frac{\left| \ln \left(\frac{1}{2} \right) \right|}{\sqrt{\pi^2 + \ln^2 \frac{1}{2}}}$
 $\leq \frac{|\ln 2|}{\sqrt{\pi^2 + \ln^2 2}} \geq \frac{\ln 2}{\sqrt{\pi^2 + \ln^2 2}}$

$\Rightarrow \boxed{\zeta \geq 0.21545}$

Hence $\theta > 77.558^\circ$

$$\zeta \omega_n > 1$$

also, $\omega_n \sqrt{1 - \zeta^2} > \pi$



③

$$\%OS = 30\%$$

$$T_p = 127s$$

$$x(t) = u(t)$$

2nd order system

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = 127 \Rightarrow \frac{1}{\omega_n \sqrt{1-\zeta^2}} = 40.42535$$

$$\%OS = \frac{3d}{100} = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} = 0.3$$

$$\Rightarrow -\frac{\zeta\pi}{\sqrt{1-\zeta^2}} = -1.2039$$

$$\Rightarrow \frac{\zeta^2}{\sqrt{1-\zeta^2}} = 0.3832$$

$$\Rightarrow \zeta^2 = 0.14687 (1-\zeta^2)$$

$$\Rightarrow 1.14687 \zeta^2 = 0.14687$$

$$\Rightarrow \zeta = 0.35786 \quad \text{--- (2)}$$

$$\text{from ① \& ②} \Rightarrow \omega_n = 0.02649$$

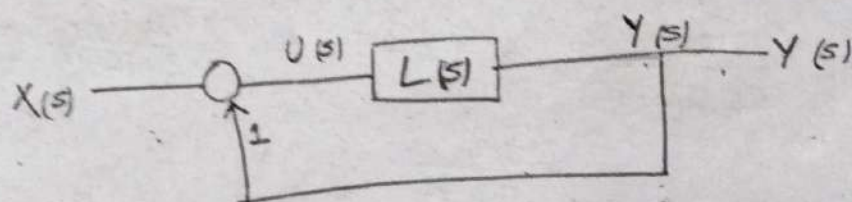
$$\text{Now } G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \boxed{\frac{7.017 \times 10^{-4}}{s^2 + 0.01836s + 7.017 \times 10^{-4}}}$$

④

Unity Feedback

Loop Transfer Function

$$L(s) = \frac{2(s+8)}{s(s+4)}$$



$$(a) \quad L(s) (U(s)) = (X(s) + Y(s)) L(s) = Y(s)$$

$$\Rightarrow X(s) L(s) = Y(s) (1 - L(s))$$

$$\Rightarrow \frac{Y(s)}{X(s)} = \frac{L(s)}{1 + L(s)}$$

$$\Rightarrow \frac{Y(s)}{X(s)} = \frac{\frac{2(s+8)}{s(s+4)}}{1 + \frac{2(s+8)}{s(s+4)}} = \frac{2(s+8)}{s^2 + 4s + 2s + 16} = \frac{2s + 16}{s^2 + 6s + 16}$$

$$\Rightarrow \boxed{G(s) = \frac{2s + 16}{s^2 + 6s + 16} = \frac{Y(s)}{R(s)}} \quad \star \star$$

$$(b) \quad R(s) = \frac{A}{s}$$

$$\Rightarrow Y(s) = \frac{A(2s+16)}{s(s^2+6s+16)} = \frac{B}{s} + \frac{Cs+D}{(s^2+6s+16)}$$

$$\Rightarrow s=0 \quad \Rightarrow B=A$$

$$\begin{aligned} \Rightarrow Cs+D &= \frac{A(2s+16)}{s} - \frac{A}{s}(s^2+6s+16) \\ &= 2A + \frac{16A}{s} - As - 6A - \frac{16A}{s} \\ &= -As - 4A \end{aligned}$$

$$\Rightarrow C = -A \quad D = -4A$$

$$Y(s) = A \left(\frac{1}{s} + \frac{-s-4}{(s+3)^2 + 7} \right)$$

$$y(t) = L^{-1}(Y(s)) = A \left(1 - e^{-3t} \cos \sqrt{7}t - \frac{1}{\sqrt{7}} e^{-3t} \sin \sqrt{7}t \right)$$

★ ★

$$Y(s) = \frac{A(2s+16)}{s^3 + 6s^2 + 16s}$$

$$y(t) = A \left(1 - e^{-3t} \cos \sqrt{7}t - \frac{1}{\sqrt{7}} e^{-3t} \sin \sqrt{7}t \right)$$

$$\frac{dy}{dt} = 3e^{-3t} \left(\cos \sqrt{7}t - \frac{1}{\sqrt{7}} \sin \sqrt{7}t \right) + e^{-3t} \left(\sqrt{7} \sin \sqrt{7}t + \cos \sqrt{7}t \right) = 0$$

$$4 \cos \sqrt{7}t + \sqrt{7} \sin \sqrt{7}t - \frac{3\sqrt{7}}{7} \sin \sqrt{7}t = 0$$

$$4 \cos \sqrt{7}t + \frac{\sin \sqrt{7}t}{\sqrt{7}} = 0$$

$$1 + \frac{1}{\sqrt{7}} \tan \sqrt{7}t = 0$$

$$\Rightarrow \tan \sqrt{7}t = -\sqrt{7}$$

$$\Rightarrow t = \frac{1}{\sqrt{7}} \tan^{-1}(-\sqrt{7})$$

$$\Rightarrow t_p = 0.00345 \text{ s}$$

$$\% \text{ OS} = 3.479 \%$$

★ ★

$$\lim_{t \rightarrow \infty} F(t) = \lim_{s \rightarrow 0} s F(s) = \lim_{s \rightarrow 0} \frac{A(2s+16)}{s(s^2 + 6s + 16)}$$

$$= \frac{A \times 16}{16} = A$$

★ ★

Steady-state
Value of $y(t)$

5.

$$A = \begin{bmatrix} -0.435 & 0.209 & 0.02 \\ 0.268 & -0.394 & 0 \\ 0.227 & 0 & -0.02 \end{bmatrix}$$

$$B = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$C = (0.0003 \quad 0 \quad 0)$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$D = 0$$

Find transfer function, UH response

T_s for X_1 .

$$\Rightarrow \frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = A X + B U$$

$$Y = C X + \cancel{D U}^0$$

$$Y = C X = 0.0003 X_1$$

$$\frac{d}{dt} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{bmatrix} -0.435 & 0.209 & 0.02 \\ 0.268 & -0.394 & 0 \\ 0.227 & 0 & -0.02 \end{bmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} U$$

$$T(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1} B + \cancel{D}^0$$

as $C = (0.003 \quad 0 \quad 0)$ & $B = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ we will only need 1 element of $(sI - A)^{-1}$

$$sI - A = \begin{bmatrix} s+0.435 & -0.209 & -0.02 \\ -0.268 & s+0.394 & 0 \\ -0.227 & 0 & s+0.02 \end{bmatrix}$$

$$\det |sI - A| = (s + 0.435)(s + 0.394)(s + 0.02) + 0.209(-0.209 \times (s + 0.02)) - 0.02(0.227(s + 0.394))$$

$$= s^3 + 0.849s^2 + 0.1274s + 0.000519$$

$$\text{1st term of } (sI - A)^{-1} = \begin{vmatrix} s + 0.394 & 0 \\ 0 & s + 0.02 \end{vmatrix}$$

$$= (s + 0.394)(s + 0.02)$$

$$\Rightarrow T(s) = \frac{0.0003 \times (s + 0.394)(s + 0.02)}{s^3 + 0.849s^2 + 0.1274s + 0.000519}$$

using residues we get, $U(s) = \frac{1}{s}$

$$Y(s) = \frac{1}{s} \frac{(0.0003)(s + 0.394)(s + 0.02)}{(s + 0.0042)(s + 0.189)(s + 0.656)}$$

$$= \frac{A}{s} + \frac{B}{s + 0.0042} + \frac{C}{s + 0.189} + \frac{D}{s + 0.656}$$

$$s = 0 \Rightarrow A = 0.00454$$

$$s = -0.656 \Rightarrow D = -0.0006$$

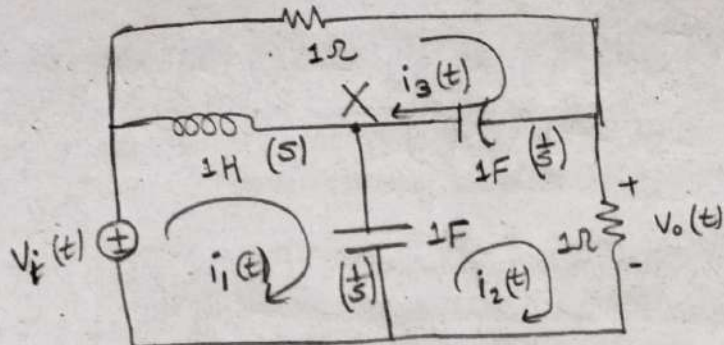
$$s = -0.0042 \Rightarrow B = -0.00365$$

$$s = -0.189 \Rightarrow C = -0.006$$

$$\Rightarrow y(t) = 0.00454 - 0.00365e^{-0.0042t} - 0.00025e^{-0.656t} - 0.0006e^{-0.189t}$$

$$(T_s)_{\max} = \frac{4}{0.0042} = 952.38 \text{ s} \quad **$$

⑥



$$\frac{q}{C} = V$$

for loop 1 $\rightarrow V_i - 1 \left(\frac{di_1}{dt} - \frac{di_3}{dt} \right)$

Using Nodal Analysis

$$\frac{X - V_i}{s} + \frac{X - V_o}{1/s} + \frac{X - 0}{1/s} = 0 \quad \text{--- (1)}$$

$$\frac{V_o - V_i}{1} + \frac{V_o - X}{1/s} + \frac{V_o}{1} = 0 \quad \text{--- (2)}$$

from (1), (2)

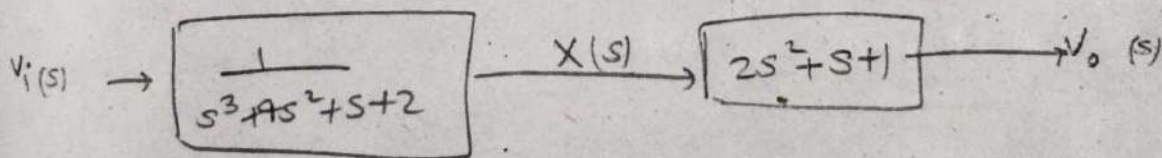
$$X \left(\frac{1}{s} + 2s \right) = \frac{V_i}{s} + sV_o$$

$$V_o - V_i + V_o + (V_o - X)s = 0$$

$$\Rightarrow X = \frac{V_i}{s} + sV_o \left/ \left(\frac{1}{s} + 2s \right) \right.$$

$$2V_o - V_i + V_o - \left(\frac{\frac{V_i}{s} + sV_o}{\frac{1}{s} + 2s} \right) s = 0$$

$$\Rightarrow G(s) = \frac{V_o(s)}{V_i(s)} = \frac{2s^2 + s + 1}{s^3 + 4s^2 + s + 2}$$



$$X_1(s) = X(s)$$

$$X_2(s) = sX(s)$$

$$X_3(s) = s^2X(s)$$

$$X(s) = \frac{1}{s^3 + 4s^2 + s + 2} V_i(s)$$

$$s^3X(s) + 4s^2X(s) + sX(s) + 2X(s) = V_i(s)$$

$$X_3 + 4X_3 + X_2 + 2X_1 = V_i(s)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} v_i$$

- 4

$$y = \begin{bmatrix} 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \text{---}$$

③

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -0.5 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & -0.5 \\ 1 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad D = 0$$

$$G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

$$sI - A = \begin{bmatrix} s+1 & 0.5 \\ -1 & s \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{\begin{bmatrix} s & 1 \\ -0.5 & s+1 \end{bmatrix}}{(s+1)s + 0.5}$$

$$C(sI - A)^{-1} = \frac{\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s & 1 \\ -0.5 & s+1 \end{bmatrix}}{s^2 + s + 0.5} = \frac{\begin{bmatrix} s & 1 \end{bmatrix}}{s^2 + s + 0.5}$$

$$C(sI - A)^{-1}B = \frac{\begin{bmatrix} s & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}}{s^2 + s + 0.5} = \frac{0.5s}{s^2 + s + 0.5}$$

**

$$\boxed{\frac{Y(s)}{U(s)} = \frac{0.5s}{s^2 + s + 0.5}}$$

Impulse response $\Rightarrow$

$$G(s) = \frac{0.5s}{s^2 + s + 0.5} = \frac{0.5s + 0.5 - 0.5}{(s+0.5)^2 + (0.5)^2}$$

$$= \boxed{0.5e^{-0.5t} \cos(0.5t) - 0.5e^{-0.5t} \sin(0.5t)}$$

**

Unit Step Response $\Rightarrow$

$$\underline{G(s)} = Y(s) = \frac{0.5s}{s(s^2+s+0.5)} = \frac{0.5}{(s+0.5)^2 + (0.5)^2}$$

$$\boxed{y(t) = e^{-0.5t} \sin(0.5t)}$$

**

Unit ramp response $\Rightarrow U(s) = \frac{1}{s^2}$

$$Y(s) = \frac{0.5}{s(s^2+s+0.5)} = \frac{A}{s} + \frac{Cs+D}{s^2+0.5s+0.5}$$

$$s=0 \Rightarrow A=1$$

$$0.5 = A(s^2+0.5s+0.5) + (Cs+D)s$$

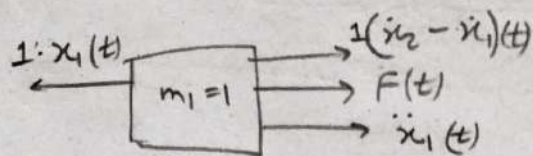
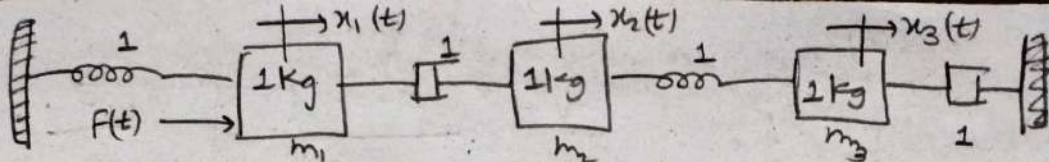
$$0.5 = s^2 + 0.5s + 0.5 + (Cs+D)s$$

$$\Rightarrow C = -1, D = -1$$

$$\Rightarrow \boxed{y(t) = e^{-0.5t} (1 - \cos(0.5t) - \sin(0.5t))}$$

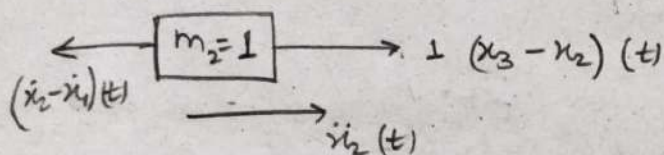
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③

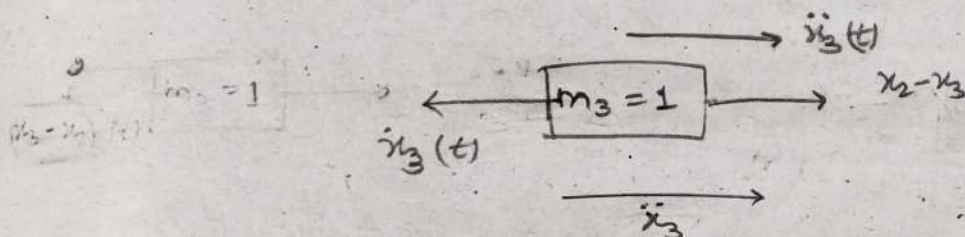


$$F(t) + \ddot{x}_2(t) - \ddot{x}_1(t) - \dot{x}_1(t) = \ddot{x}_1(t)$$

$$\Rightarrow F(t) = \dot{x}_1(t) + \ddot{x}_1(t) - \ddot{x}_2(t) + \ddot{x}_1(t) \quad \text{--- ①}$$



$$x_3(t) - x_2(t) - \ddot{x}_2(t) + \dot{x}_1(t) = \ddot{x}_2(t) \quad \text{--- ②}$$



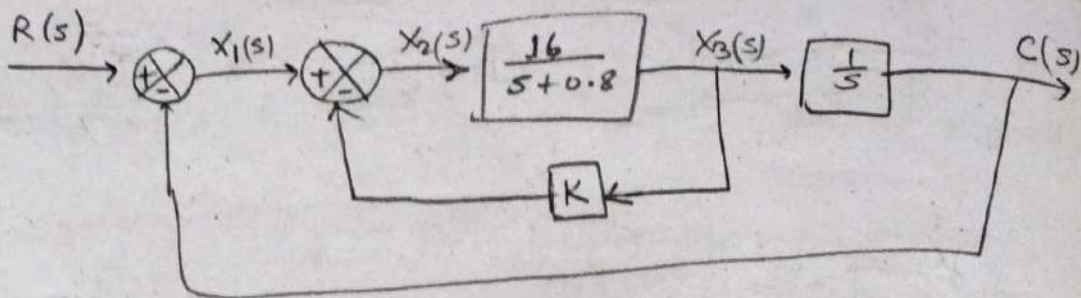
$$x_2 - x_3 - \ddot{x}_3 = \ddot{x}_3 \Rightarrow \ddot{x}_3 = x_2 - x_3 - \ddot{x}_3 \quad \text{--- ③}$$

$$\begin{pmatrix} \dot{x}_1 \\ \ddot{x}_1 \\ \dot{x}_2 \\ \ddot{x}_2 \\ \dot{x}_3 \\ \ddot{x}_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ \dot{x}_1 \\ x_2 \\ \dot{x}_2 \\ x_3 \\ \dot{x}_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} F(t)$$

$$y = x_3(t)$$

★★★

9



$$R(s) - C(s) = X_1(s)$$

$$X_1(s) - K X_3(s) = X_2(s)$$

$$X_2(s) \frac{16}{s+0.8} = X_3(s)$$

$$\frac{X_3(s)}{s} = C(s)$$

$$R(s) - C(s) = X_2(s) + K X_3(s) = \frac{s+0.8}{16} X_3(s) + K X_3(s)$$

$$= \left(\frac{s+0.8}{16} + K \right) X_3(s) = \left(\frac{s+0.8}{16} + K \right) s C(s)$$

$$R(s) = \left(\frac{s^2 + 0.8s + 16Ks}{16} \right) C(s) + C(s)$$

$$= \frac{s^2 + (0.8 + 16K)s + 16}{16} C(s)$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{16}{s^2 + (0.8 + 16K)s + 16} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n = 4$$

$$2\zeta\omega_n = 0.8 + 16K = 2\zeta + = 8\zeta$$

$$\zeta = 0.1 + 2K$$

Damping

$$\text{ratio} = \zeta = 0.5$$

$$\Rightarrow 0.5 = 0.1 + 2K$$

$$\Rightarrow K = \frac{0.4}{2} = 0.2$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{16}{s^2 + 4s + 16} = \frac{16}{(s+2)^2 + 12}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = \frac{\pi}{4\sqrt{1-(0.5)^2}} = \frac{\pi}{4\sqrt{0.75}} = 0.9069 \text{ s}$$

$$T_s = \frac{4}{\zeta \omega_n} = \frac{4}{4(0.5)} = 2$$

$$\begin{aligned} \phi.05 &= 100 \times e^{-\frac{\zeta \pi}{\sqrt{1-\zeta^2}}} = 100 \times e^{-\frac{0.5\pi}{\sqrt{0.75}}} \\ &= 100 \times e^{-\frac{\pi}{\sqrt{3}}} = 16.30 \% \end{aligned}$$

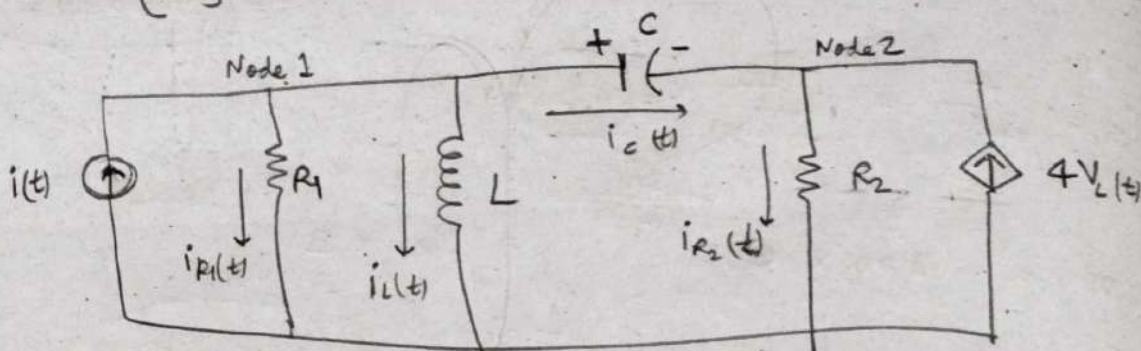
Rise Time (10% $\rightarrow$ 90%)

From book, for $\zeta = 0.5$ $T_r = 1.638$
for $\omega_n = 1$ 4 $T_r \propto \frac{1}{\omega_n}$

$$\Rightarrow \omega_n = 4 \Rightarrow T_r = \frac{1.638}{4} = 0.4095 \text{ s}$$

10

$$y = \begin{bmatrix} v_{R_2} \\ i_{R_2} \end{bmatrix}$$



States $\Rightarrow i_L$ and v_C

$$i(t) = i_{R_1}(t) + i_L(t) + i_C(t) = i_{R_1} + i_L + C \frac{dv_C}{dt} \quad \text{--- (1)}$$

$$i_{R_2} = i_C + 4v_L = C \frac{dv_C}{dt} + 4L \frac{di_L}{dt} \quad \text{--- (3)}$$

$$i_{R_1} R_1 = L \frac{di_L}{dt} \quad \text{--- (2)}$$

①, ② $\Rightarrow$

$$i(t) = \frac{L}{R_1} \frac{di_L}{dt} + i_L + C \frac{dv_C}{dt} \quad \text{--- (5)}$$

$$i_{R_1} R_1 - v_C - i_{R_2} R_2 = 0 \quad \text{--- (4)}$$

$$L \frac{di_L}{dt} - v_C = i_{R_2} R_2$$

③, ④ $\Rightarrow$

$$L \frac{di_L}{dt} - v_C = C R_2 \frac{dv_C}{dt} + 4L R_2 \frac{di_L}{dt} \quad \text{--- (6)}$$

$$\text{⑤, ⑥} \Rightarrow C \frac{dv_C}{dt} = \frac{i - i_L - \frac{L}{R_1} \frac{di_L}{dt}}{1}$$

$$L \frac{di_L}{dt} - v_C = R_2 \left(i - i_L - \frac{L}{R_1} \frac{di_L}{dt} \right) + 4L R_2 \frac{di_L}{dt}$$

$$\boxed{L \frac{di_L}{dt} - v_C = R_2 i - R_2 i_L - \frac{R_2 L}{R_1} \frac{di_L}{dt} + 4L R_2 \frac{di_L}{dt}}$$

$$(4LR_2 + L + \frac{R_2 L}{R_1}) \frac{di_L}{dt} = V_c + R_2 i - R_2 i_L$$

$$\Rightarrow \frac{di_L}{dt} = \frac{V_c + R_2 i - R_2 i_L}{(L + \frac{R_2 L}{R_1} - 4LR_2)} \Rightarrow \textcircled{F_1}$$

$$L + \frac{R_2 L}{R_1} - 4LR_2 = K$$

$$\Rightarrow \frac{di_L}{dt} = \frac{V_c + R_2 i - R_2 i_L}{K}$$

$$\begin{aligned} C \frac{dV_c}{dt} &= i - i_L - \frac{L}{R_1} \times \frac{(V_c + R_2 i - R_2 i_L)}{K} \\ &= i \left(1 + \frac{LR_2}{RK}\right) + i_L \left(-1 + \frac{LR_2}{RK}\right) - \frac{L}{RK} V_c \end{aligned}$$

$$\Rightarrow \frac{dV_c}{dt} = \left(\frac{1 - \frac{LR_2}{RK}}{C}\right) i + i_L \left(\frac{\frac{LR_2}{RK} - 1}{C}\right) - \frac{L}{RKC} V_c \Rightarrow \textcircled{F_2}$$

$$\Rightarrow \begin{pmatrix} \frac{di_L}{dt} \\ \frac{dV_c}{dt} \end{pmatrix} = \begin{pmatrix} \frac{1}{K} & -\frac{R_2}{K} \\ -\frac{L}{RKC} & \frac{\frac{LR_2}{RK} - 1}{C} \end{pmatrix} \begin{pmatrix} V_c \\ i_L \end{pmatrix} + \begin{pmatrix} \frac{R_2}{K} \\ \frac{1 - \frac{LR_2}{RK}}{C} \end{pmatrix} i(t)$$

from ③ $i_{R_2} = C \frac{dV_c}{dt} + 4L \frac{di_L}{dt}$

from ④ $i_{R_2} = \frac{1}{R_2} \left(L \frac{di_L}{dt} - V_c \right)$

$$i_{R_2} = \frac{L}{R_2} K (V_c + R_2 i - R_2 i_L) - \frac{V_c}{R_2} \Rightarrow \textcircled{F_3}$$

$$V_{R_2} = i_{R_2} R_2 = \frac{L}{K} (V_c + R_2 i - R_2 i_L) - \frac{V_c}{1} \Rightarrow \textcircled{F_4}$$

$$\Rightarrow \begin{pmatrix} v_{R_2} \\ i_{R_2} \end{pmatrix} = \begin{pmatrix} \frac{L}{K} - 1 & -\frac{R_2 L}{K} \\ \frac{L}{R_2 K} - \frac{1}{R_2} & -\frac{L}{K} \end{pmatrix} \begin{pmatrix} v_c \\ i_L \end{pmatrix} + \begin{pmatrix} \frac{L R_2}{K} \\ \frac{1}{K} \end{pmatrix} i(t)$$
